Algebraic Structures

Module-3

Algebraic Structures: Groups – Cyclic groups, Cosets, Modulo groups - Primitive roots - Discrete logarithms. Rings – Sub rings, ideals and quotient rings, Integral domains. Fields – Finite fields . Classification - Structure of finite fields. Lattice, Lattice as Algebraic system, sub lattices, some special lattices.

Definitions and Examples

Binary operation $(*)$

A binary operation $*$ on a set S is a function mapping $S \times S$ into S.

For each $(a, b) \in S \times S$, we will denote the element $*((a, b))$ of S by $a * b$.

$$\text{i.e. } *(a, b) = a * b$$

We may regard a binary operation $*$ on S as assigning, to each ordered pair (a, b) of elements of S, an element $a * b$ of S.

Ex: usual addition $+$ on $\mathbb{R}$, usual multiplication $\cdot$ on $\mathbb{R}$.

Example On $\mathbb{Z}^+$, we define an operation $*$ by $a * b$ equals the smaller of a and b, or the common value if $a = b$. Thus $2 * 11 = 2; 15 * 10 = 10$; and $3 * 3 = 3$. ▲

Example On $\mathbb{Z}^+$, we define an operation $*'$ by $a *' b = a$. Thus $2 *' 3 = 2; 25 *' 10 = 25$; and $5 *' 5 = 5$. ▲

Note that we require an operation on a set S to be defined for *every* ordered pair (a, b) of elements from S.

Let $M(\mathbb{R})$ be the set of all matrices with real entries.

The usual matrix addition $+$ is *not* an operation on this set since $A + B$ is not defined for an ordered pair (A, B) of matrices having different numbers of rows or of columns.

- A set S is closed under $*$, a binary operation, but a subset may not be.
- Let $*$ be an operation on S and let H be a subset of S. The subset H is **closed under** $*$ if for all $a, b \in H$ we also have $a * b \in H$.

Example Our usual addition $+$ on the set $\mathbb{R}$ of real numbers does not induce an operation on the set $\mathbb{R}^*$ of nonzero real numbers because $2 \in \mathbb{R}^*$ and $-2 \in \mathbb{R}^*$, but $2 + (-2) = 0$ and $0 \notin \mathbb{R}^*$. Thus $\mathbb{R}^*$ is not closed under $*$. ▲

Let $+$ and $\cdot$ be the usual operations of addition and multiplication on the set $\mathbb{Z}$, and let $H = \{n^2 | n \in \mathbb{Z}^+\}$. Determine whether H is closed under (a) addition and (b) multiplication.

For part (a), we need only observe that $1^2 = 1$ and $2^2 = 4$ are in H, but that $1 + 4 = 5$ and $5 \notin H$. Thus H is not closed under addition.

For part (b), suppose that $r \in H$ and $s \in H$. Using what it means for r and s to be in H, we see that there must be integers n and m in $\mathbb{Z}^+$ such that $r = n^2$ and $s = m^2$. Consequently, $rs = n^2m^2 = (nm)^2$. By the characterization of elements in H and the fact that $nm \in \mathbb{Z}^+$, this means that $rs \in H$, so H is closed under multiplication. ▲

Let F be the set of all real-valued functions f having as domain the set $\mathbb{R}$ of real numbers. We are familiar from calculus with the operations $+, -, \cdot$, and $\circ$ on F. Namely, for each ordered pair (f, g) of functions in F, we define for each $x \in \mathbb{R}$

$$\begin{aligned} f+g \text{ by } (f+g)(x) &= f(x)+g(x) && \text{addition,} \\ f-g \text{ by } (f-g)(x) &= f(x)-g(x) && \text{subtraction,} \\ f \cdot g \text{ by } (f \cdot g)(x) &= f(x)g(x) && \text{multiplication, and} \\ f \circ g \text{ by } (f \circ g)(x) &= f(g(x)) && \text{composition.} \end{aligned}$$

All four of these functions are again real valued with domain $\mathbb{R}$, so F is closed under all four operations $+, -, \cdot$, and $\circ$. ▲

- An operation $*$ on a set S is **commutative** if (and only if) $a * b = b * a$ for all $a, b \in S$.
- An operation on a set S is **associative** if $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$.
- Let S be a set with binary operation $*$. If $e \in S$ has the property that for all $a \in S$, $a * e = e * a = a$, then e is called an **identity element** for $*$. A set with binary operation $*$ has at most one identity element.

The numbers 0 and 1 play special roles as real numbers because for any real number a,
$a + 0 = a$ and $a \times 1 = a$.

0 is called the *additive identity* in $\mathbb{R}$ and 1 is called the *multiplicative identity* in $\mathbb{R}$.

- If $*$ is an operation on the set S and S has an identity e, then for any $x \in S$,

the **inverse** of x is an element x' such that $x * x' = x' * x = e$.

For addition, the inverse of a real number a is $-a$.

Using multiplication, the inverse of a nonzero real number a is $\frac{1}{a}$.

Group

Definition A **group** $\langle G, * \rangle$ is a set G, closed under a binary operation $*$, such that the following axioms are satisfied:

$\mathscr{G}_1$: For all $a, b, c \in G$, we have

$$(a * b) * c = a * (b * c). \quad \textbf{associativity of } *$$

$\mathscr{G}_2$: There is an element e in G such that for all $x \in G$,

$$e * x = x * e = x. \quad \textbf{identity element } e \text{ for } *$$

$\mathscr{G}_3$: Corresponding to each $a \in G$, there is an element a' in G such that

$$a * a' = a' * a = e. \quad \textbf{inverse } a' \text{ of } a$$ ■

A group G is **abelian** if its binary operation is commutative.

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$\mathscr{G}_3$: Corresponding to each $a \in G$, there is an element a' in G such that

$$a * a' = a' * a = e. \quad \textbf{inverse } a' \text{ of } a$$

■

$< \mathbb{R}$,+> is a group with identity element 0 and the inverse of any real number a is $-a$.

$< \mathbb{R}, \cdot >$ is not a group since 0 has no multiplicative inverse.

- A group $< G,* >$ is a finite group if G is a finite set.
- The order of a finite group $< G,* >$, denoted by $ord(G)$ or $O(G)$, is equal to $|G|$.
- If the group operation is multiplication, then the identity is usually denoted by $1\ or\ e$ and the inverse of a by a^{-1}.
- If the group operation is addition, then the identity is usually denoted by 0 and the inverse of a by $-a$.

Question:

Show that the subset S of $M_n(\mathbb{R})$ consisting of all *invertible* $n \times n$ matrices under matrix multiplication is a group.

Solution We start by showing that S is closed under matrix multiplication. Let A and B be in S, so that both A^{-1} and B^{-1} exist and $AA^{-1} = BB^{-1} = I_n$. Then

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = I_n,$$

so that AB is invertible and consequently is also in S.

Since matrix multiplication is associative and I_n acts as the identity element, and since each element of S has an inverse by definition of S, we see that S is indeed a group. This group is *not* commutative. ▲

Question:

Is $< \mathbb{Q}^+, * >$ a group where $*$ is defined by $a * b = \frac{ab}{2}$?

Let $*$ be defined on $\mathbb{Q}^+$ by $a * b = ab/2$. Then

$$(a * b) * c = \frac{ab}{2} * c = \frac{abc}{4},$$

and likewise

$$a * (b * c) = a * \frac{bc}{2} = \frac{abc}{4}.$$

Thus $*$ is associative. Computation shows that

$$2 * a = a * 2 = a$$

for all $a \in \mathbb{Q}^+$, so 2 is an identity element for $*$. Finally,

$$a * \frac{4}{a} = \frac{4}{a} * a = 2,$$

so $a' = 4/a$ is an inverse for a. Hence $\mathbb{Q}^+$ with the operation $*$ is a group. ▲

More examples:

The set $\mathbb{Z}^+$ under addition is *not* a group. There is no identity element for $+$ in $\mathbb{Z}^+$. ▲

The set of all nonnegative integers (including 0) under addition is still *not* a group. There is an identity element 0, but no inverse for 2. ▲

The familiar additive properties of integers and of rational, real, and complex numbers show that $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ under addition are abelian groups. ▲

The set $\mathbb{Z}^+$ under multiplication is *not* a group. There is an identity 1, but no inverse of 3. ▲

The familiar multiplicative properties of rational, real, and complex numbers show that the sets $\mathbb{Q}^+$ and $\mathbb{R}^+$ of positive numbers and the sets $\mathbb{Q}^*$, $\mathbb{R}^*$, and $\mathbb{C}^*$ of nonzero numbers under multiplication are abelian groups. ▲

The set of all real-valued functions with domain $\mathbb{R}$ under function addition is a group. This group is abelian. ▲

The set $M_{m\times n}(\mathbb{R})$ of all $m \times n$ matrices under matrix addition is a group. The $m \times n$ matrix with all entries 0 is the identity matrix. This group is abelian. ▲

The set $M_n(\mathbb{R})$ of all $n \times n$ matrices under matrix multiplication is *not* a group. The $n \times n$ matrix with all entries 0 has no inverse. ▲

Example **(the additive group $\mathbb{Z}_n$)** Let $n \geq 2$ be an integer. Then $(\mathbb{Z}_n, +)$ is a finite abelian group of order n, where $+$ denotes addition modulo n. The identity element is 0, and the inverse of a, usually denoted $-a$, is $(-a) \bmod n$.

Remark Suppose $1 \leq a < n$. Then $(-a) \bmod n = n - a$.

Example The additive inverses of the elements in $(\mathbb{Z}_{10}, +)$ are as follows: $-0 = 0$, $-1 = 9$, $-2 = 8$, $-3 = 7$, $-4 = 6$, $-5 = 5$, $-6 = 4$, $-7 = 3$, $-8 = 2$, and $-9 = 1$.

Example **(the multiplicative group $\mathbb{Z}_p{}^*$)** Let $p \geq 2$ be a prime. Define $\mathbb{Z}_p{}^* = \mathbb{Z}_p \backslash \{0\}$. Then $(\mathbb{Z}_p{}^*, \cdot)$ is a finite abelian group of order $p-1$, where $\cdot$ denotes multiplication modulo p. The identity element is 1, and the inverse of a, usually denoted a^{-1}. $\mathbb{Z}_p^* = \{1, 2, \cdots, p-1\}$

The order of $(\mathbb{Z}_n{}^*, \cdot)$ is equal to $\phi(n)$.

Example The multiplicative inverses of the elements in $(\mathbb{Z}_{11}{}^*, \cdot)$ are as follows: $1^{-1} = 1$, $2^{-1} = 6$, $3^{-1} = 4$, $4^{-1} = 3$, $5^{-1} = 9$ $6^{-1} = 2$, $7^{-1} = 8$, $8^{-1} = 7$, $9^{-1} = 5$, and $10^{-1} = 10$.

Remark To verify that $a^{-1} \bmod n = b$ in $\mathbb{Z}_n{}^*$, it is sufficient to check that $ab - 1$ is divisible by n.

Question: What are the elements of the multiplicative group $< \mathbb{Z}_{20}^*, \cdot >$? What is the order of this group? Also, find the multiplicative inverses of the elements.

The order of $(\mathbb{Z}_{20}{}^*, \cdot)$ is equal to $\phi(20) = (2^2 - 2)(5 - 1) = 8$.

The elements in $(\mathbb{Z}_{20}{}^*, \cdot)$ are $1, 3, 7, 9, 11, 13, 17$, and 19.

The multiplicative inverses are as follows:

$1^{-1} = 1, 3^{-1} = 7, 7^{-1} = 3, 9^{-1} = 9, 11^{-1} = 11, 13^{-1} = 17, 17^{-1} = 13,$ and $19^{-1} = 19$.

Subgroup

If a subset H of a group G is closed under the binary operation of G and if H with the induced operation from G is itself a group, then H is a **subgroup of** G. We shall let $H \leq G$ or $G \geq H$ denote that H is a subgroup of G, and $H < G$ or $G > H$ shall mean $H \leq G$ but $H \neq G$. ■

Theorem A subset H of a group G is a subgroup of G if and only if

1. H is closed under the binary operation of G,
2. the identity element e of G is in H, and
3. for all $a \in H$, $a^{-1} \in H$ also.

Subgroup

If a subset H of a group G is closed under the binary operation of G and if H with the induced operation from G is itself a group, then H is a **subgroup of** G. We shall let $H \leq G$ or $G \geq H$ denote that H is a subgroup of G, and $H < G$ or $G > H$ shall mean $H \leq G$ but $H \neq G$. ■

Thus $\langle \mathbb{Z}, + \rangle < \langle \mathbb{R}, + \rangle$ but $\langle \mathbb{Q}^+, \cdot \rangle$ is *not* a subgroup of $\langle \mathbb{R}, + \rangle$, even though as sets, $\mathbb{Q}^+ \subset \mathbb{R}$. Every group G has as subgroups G itself and $\{e\}$, where e is the identity element of G.

If G is a group, then the subgroup consisting of G itself is the **improper subgroup** of G. All other subgroups are **proper subgroups**. The subgroup $\{e\}$ is the **trivial subgroup** of G. All other subgroups are **nontrivial**. ■

Examples:

Let $\mathbb{R}^n$ be the additive group of all n-component row vectors with real number entries. The subset consisting of all of these vectors having 0 as entry in the first component is a subgroup of $\mathbb{R}^n$. ▲

$\mathbb{Q}^+$ under multiplication is a proper subgroup of $\mathbb{R}^+$ under multiplication. ▲

Orders of Group Elements

For a finite group $(G,*)$, define the order of an element of an element $a \in G$, denoted by $ord(a)$, to be the smallest positive integer m such that

$$a * a * a \cdots * a \ (m \ times) = e, \text{ the identity element.}$$

In other words, $ord(a)$ is the smallest positive integer m such that $a^m = e$ (in addition notation, we write $ma = e$).

- The identity element is defined to have order 1.
- Any nonidentity element has order greater than 1.
- If no such integer exists, we say that a has infinite order.

To find the order of a group element g, you need only compute the sequence of products $g, g^2, g^3, \cdots$, until you reach the identity for the first time.

The exponent of this product (or coefficient if the operation is addition) is the order of g.

If the identity never appears in the sequence, then g has infinite order.

Consider $\mathbb{Z}_{10}$under addition modulo 10.

- $1.2 = 2,$
- $2.2 = 2 + 2 = 4,$
- $3.2 = 2 + 2 + 2 = 6,$
- $4.2 = 2 + 2 + 2 + 2 = 8,$
- $5.2 = 2 + 2 + 2 + 2 + 2 = 0 \Rightarrow ord(2) = 5$

Similar computations show that

$$ord(0) = 1,$$

$$ord(7) = 10,$$

$$ord(5) = 2,$$

$$ord(6) = 5.$$

Consider $\mathbb{Z}$ under ordinary addition.

Here every nonzero element has infinite order, since the sequence $a, 2a, 3a, \cdots$ never includes 0 when $a \neq 0$.

Results:

- For a finite group $(G,*)$, the order of any $a \in G$ divides the order of the group, i.e., $\boldsymbol{ord}(a)|\boldsymbol{ord}(G)$.
- In a group of prime order p, every nonidentity element has order p.

Cyclic Groups

A group G is called **cyclic**

if there is an element a in G such that $G = \{a^n : n \in \mathbb{Z}\}$.

Such an element a is called a generator of G. We write $G = < a >$.

■ **EXAMPLE 1** The set of integers Z under ordinary addition is cyclic. Both 1 and -1 are generators. (Recall that, when the operation is addition, 1^n is interpreted as

$$\underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}}$$

when n is positive and as

$$\underbrace{(-1) + (-1) + \cdots + (-1)}_{|n| \text{ terms}}$$

when n is negative.) ■

EXAMPLE 2 The set $Z_n = \{0, 1, \ldots, n-1\}$ for $n \geq 1$ is a cyclic group under addition modulo n. Again, 1 and $-1 = n - 1$ are generators.

Unlike Z, which has only two generators, Z_n may have many generators (depending on which n we are given).

EXAMPLE 3 $Z_8 = \langle 1 \rangle = \langle 3 \rangle = \langle 5 \rangle = \langle 7 \rangle$. To verify, for instance, that $Z_8 = \langle 3 \rangle$, we note that $\langle 3 \rangle = \{3, 3+3, 3+3+3, \ldots\}$ is the set $\{3, 6, 1, 4, 7, 2, 5, 0\} = Z_8$. Thus, 3 is a generator of Z_8. On the other hand, 2 is not a generator, since $\langle 2 \rangle = \{0, 2, 4, 6\} \neq Z_8$.

Example Let $n \geq 2$ be an integer. Then $(\mathbb{Z}_n, +)$ is a cyclic group, and 1 is a generator.

Further, an element $a \in \mathbb{Z}_n$ is a generator of $(\mathbb{Z}_n, +)$ if and only if $\gcd(a, n) = 1$.

The number of generators of $(\mathbb{Z}_n, +)$ is $\phi(n)$.

Example Let $p \geq 2$ be a prime. Then $(\mathbb{Z}_p{}^*, \cdot)$ is a cyclic group of order $p-1$, and a generator of this group is called a ***primitive element modulo*** p.

Theorem *$(\mathbb{Z}_n{}^*, \cdot)$ is a cyclic group (of order $\phi(n)$) if and only if $n = 2, 4, p^k$ or $2p^k$, where p is an odd prime and k is a positive integer.*

Primitive root

Let $p \geq 2$ be a prime. Then $(\mathbb{Z}_p^*, \cdot)$ is a cyclic group of order $p-1$, and a generator of this group is called a ***primitive element modulo*** p.

> *$\alpha \in \mathbb{Z}_p^*$ is a primitive element if and only if*
>
> $$\alpha^{(p-1)/q} \not\equiv 1 \pmod{p}$$
>
> *for all primes q such that $q|(p-1)$.*

The number of primitive elements in $(\mathbb{Z}_p^*, \cdot)$ is $\phi(p-1)$

Consider $(\mathbb{Z}_{13}^*, \cdot)$ where $\mathbb{Z}_{13}^* = \{1, 2, 3, \ldots, 12\}$.

Then $\alpha \in \mathbb{Z}_{13}^*$ is a primitive root (element) of $\mathbb{Z}_{13}^*$ iff $\alpha^{12/q} \not\equiv 1 \ (mod\ 13)$ for all primes q such that $q \mid 12$.

Here, $p - 1 = 12 = 2^2 3^1$. Hence, the possible values of q are $2, 3$.

Therefore, to verify that $\alpha = 2$ is a primitive element of $\mathbb{Z}_{13}^*$ it is sufficient to check that $2^6 \not\equiv 1 \ (mod\ 13)$ and $2^4 \not\equiv 1 \ (mod\ 13)$.

The number of primitive elements of $\mathbb{Z}_{13}^*$ is $\phi(12) = 12\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 4$.

Cosets

Suppose $(G,*)$ is a finite group and H is a subgroup of G.

For any $a \in G$, define the **right coset Ha** as follows:

$$Ha = \{h * a : h \in H\}.$$

For any $a \in G$, define the **left coset aH** as follows:

$$aH = \{a * h : h \in H\}.$$

A group G can be partitioned into right (or left) cosets of any subgroup H.

Example Consider the group $G = (\mathbb{Z}_{19}{}^*, \cdot)$, which is a cyclic group of order 18. It can be verified that 2 is a primitive element in G. The element $2^3 = 8$ generates a subgroup H of order $18/3 = 6$, where

$$H = \{1, 8, 7, 18, 11, 12\}.$$

There are two additional cosets of H, namely

$$2H = \{2, 16, 14, 17, 3, 5\}$$

and

$$4H = \{4, 13, 9, 15, 6, 10\}.$$

Question: Find all the left coset of the subgroup H $=< 3 >$ of $\mathbb{Z}_{12}$.

One coset is always the subgroup itself, so $0 + H = \{0, 3, 6, 9\}$.

We next find $1 + H = \{1, 4, 7, 10\}$.

Finally, $2 + H = \{2, 5, 8, 11\}$.

Question: Exhibit the left coset of the subgroup $3\mathbb{Z}$ of $\mathbb{Z}$.

Solution Our notation here is additive, so the left coset of $3\mathbb{Z}$ containing m is $m + 3\mathbb{Z}$. Taking $m = 0$, we see that

$$3\mathbb{Z} = \{\cdots, -9, -6, -3, 0, 3, 6, 9, \cdots\}$$

is itself one of its left cosets, the coset containing 0. To find another left coset, we select an element of $\mathbb{Z}$ not in $3\mathbb{Z}$, say 1, and find the left coset containing it. We have

$$1 + 3\mathbb{Z} = \{\cdots, -8, -5, -2, 1, 4, 7, 10, \cdots\}.$$

These two left cosets, $3\mathbb{Z}$ and $1 + 3\mathbb{Z}$, do not yet exhaust $\mathbb{Z}$. For example, 2 is in neither of them. The left coset containing 2 is

$$2 + 3\mathbb{Z} = \{\cdots, -7, -4, -1, 2, 5, 8, 11, \cdots\}.$$

It is clear that these three left cosets we have found do exhaust $\mathbb{Z}$, so they constitute the partition of $\mathbb{Z}$ into left cosets of $3\mathbb{Z}$. ▲

Lagrange's Theorem

Let H be a subgroup of a finite group G. Then the order of H is a divisor of the order of G.

Discrete Logarithm

The Discrete Logarithm problem forms the basis of numerous cryptographic protocols.

Problem : Discrete Logarithm

Instance: A multiplicative group $(G, \cdot)$, an element $\alpha \in G$ having order n, and an element $\beta \in \langle \alpha \rangle$.

Question: Find the unique integer a, $0 \leq a \leq n-1$, such that

$$\alpha^a = \beta.$$

We will denote this integer a by $\log_\alpha \beta$; it is called the ***discrete logarithm*** of β.

The ElGamal Cryptosystem

- ElGamal proposed a public-key cryptosystem that is based on the **Discrete Logarithm** problem in$(\mathbb{Z}_p^*, \cdot)$.
- The encryption operation in the ElGamal Cryptosystem is randomized, since the ciphertext depends on both the plaintext x and on the random value k chosen by the sender.
- Hence, there will be many ciphertexts ($p - 1$, in fact) that are encryptions of the same plaintext.

Definition :

A ***cryptosystem*** is a five-tuple $(\mathcal{P}, \mathcal{C}, \mathcal{K}, \mathcal{E}, \mathcal{D})$, where

1. $\mathcal{P}$ is a finite set of possible ***plaintexts***;
2. $\mathcal{C}$ is a finite set of possible ***ciphertexts***;
3. $\mathcal{K}$, the ***keyspace***, is a finite set of possible ***keys***;
4. For each $K \in \mathcal{K}$, there is an ***encryption rule*** $e_K \in \mathcal{E}$ and a corresponding ***decryption rule*** $d_K \in \mathcal{D}$. Each $e_K : \mathcal{P} \to \mathcal{C}$ and $d_K : \mathcal{C} \to \mathcal{P}$ are functions such that $d_K(e_K(x)) = x$ for every plaintext element $x \in \mathcal{P}$.

Cryptosystem : *ElGamal Public-key Cryptosystem in* $\mathbb{Z}_p{}^*$

Let p be a prime such that the **Discrete Logarithm** problem in $(\mathbb{Z}_p{}^*, \cdot)$ is infeasible, and let $\alpha \in \mathbb{Z}_p{}^*$ be a primitive element. Let $\mathcal{P} = \mathbb{Z}_p{}^*$, $\mathcal{C} = \mathbb{Z}_p{}^* \times \mathbb{Z}_p{}^*$, and define

$$\mathcal{K} = \{(p, \alpha, a, \beta) : \beta \equiv \alpha^a \pmod{p}\}.$$

The values p, α, and β are the public key, and a is the private key.

For $K = (p, \alpha, a, \beta)$, and for a (secret) random number $k \in \mathbb{Z}_{p-1}$, define

$$e_K(x, k) = (y_1, y_2),$$

where

$$y_1 = \alpha^k \bmod p$$

and

$$y_2 = x\beta^k \bmod p.$$

For $y_1, y_2 \in \mathbb{Z}_p{}^*$, define

$$d_K(y_1, y_2) = y_2({y_1}^a)^{-1} \bmod p.$$

Example Suppose $p = 2579$ and $\alpha = 2$. It can be verified that α is a primitive element modulo p. Let $a = 765$, so

$$\beta = 2^{765} \bmod 2579 = 949.$$

Now, suppose that Alice wishes to send the message $x = 1299$ to Bob. Say $k = 853$ is the random integer she chooses. Then she computes

$$\begin{aligned} y_1 &= 2^{853} \bmod 2579 \\ &= 435, \end{aligned}$$

and

$$\begin{aligned} y_2 &= 1299 \times 949^{853} \bmod 2579 \\ &= 2396. \end{aligned}$$

When Bob receives the ciphertext $y = (435, 2396)$, he computes

$$\begin{aligned} x &= 2396 \times (435^{765})^{-1} \bmod 2579 \\ &= 1299, \end{aligned}$$

which was the plaintext that Alice encrypted. □

- Clearly the ElGamal Cryptosystem will be insecure if a third person can compute the value $a = \log_\alpha \beta$, for then he can decrypt ciphertexts exactly as Bob does.
- Hence, a necessary condition for the ElGamal Cryptosystem to be secure is that the **Discrete Logarithm** problem in $\mathbb{Z}_p^*$ is computationally hard.
- For a secure setting, it is recommended that p should have at least 2048 bits in its binary representation, and $p - 1$ should have at least one "large" prime factor.

Rings

A ring is a triple $R = (X, \cdot, +)$, where X is a finite set and $\cdot$ and $+$ are binary operations defined on X, that satisfies the following properties:

- $(X, +)$ is an abelian group with identity 0.
- Multiplication is ***associative***, i.e., for any $a, b, c \in X$, $(ab)c = a(bc)$.
- The ***distributive property*** is satisfied, i.e., for any $a, b, c \in X$, $(a + b)c = (ac) + (bc)$ and $a(b + c) = (ab) + (ac)$.

$\mathbb{Z}, \mathbb{R}, \mathbb{C}$ are all infinite rings

A ring $R = (X, \cdot, +)$ is a ***finite ring*** if X is a finite set.

A ring $R = (X, \cdot, +)$ is a ***ring with identity*** if X contains a multi plicative identity, denoted by 1.

A ring $R = (X, \cdot, +)$ is a ***commutative ring*** if multiplication is commutative.

Example $(\mathbb{Z}_m, \cdot, +)$ is a finite ring for any $m \geq 2$.

Example **(matrices)** Let $n \geq 2$. The set of $n \times n$ matrices with entries from $\mathbb{Z}_p$ is a ring, but not a commutative ring.

Example $\mathbb{Z}_2[x]$ denotes the ring of polynomials with coefficients from $\mathbb{Z}_2$. We add and multiply polynomials in the usual way, but we reduce the coefficients modulo 2. For example, if $a(x) = x^2 + 1$ and $b(x) = x^2 + x + 1$, we would have $a(x) + b(x) = x$ and $a(x)b(x) = x^4 + x^3 + x + 1$.

Ideals and Quotient Rings

Definition (ideal) Suppose $R = (X, \cdot, +)$ is a commutative ring. An ***ideal*** is a subset $I \subseteq X$ that satisfies the following properties:

- $(I, +)$ is an abelian group, and
- $ab \in I$ whenever $a \in X$ and $b \in I$.

Definition (principal ideal) Suppose $R = (X, \cdot, +)$ is a commutative ring and let $c \in X$. The ***principal ideal*** generated by c, which is denoted by (c), is the subset defined as follows:

$$(c) = \{ac : a \in X\}.$$

It is easy to see that a principal ideal is always an ideal.

Definition (quotient ring) Suppose $R = (X, \cdot, +)$ is a commutative ring and $I = (c)$ is a principal ideal. The ***quotient ring*** R/I is constructed as follows. $R/I = (Y, \cdot, +)$, where Y consists of the (additive) cosets of I in $(X, +)$. The sum of two cosets $I + a$ and $I + b$ is defined to be $I + (a + b)$, for any $a, b \in X$, and the product of the two cosets $I + a$ and $I + b$ is defined to be $I + ab$.

Integral Domain

- A **zero-divisor** is a nonzero element a of a commutative ring R such that there is a nonzero element $b \in R$ with $ab = 0$.
- An **integral domain** is a commutative ring with unity and no zero-divisors.
- Thus, in an integral domain, a product is 0 only when one of the factors is 0; that is, $ab = 0$ only when $a = 0$ or $b = 0$.

■ **EXAMPLE 1** The ring of integers is an integral domain. ■

■ **EXAMPLE 2** The ring of Gaussian integers $Z[i] = \{a+bi \mid a, b \in Z\}$ is an integral domain. ■

■ **EXAMPLE 3** The ring $Z[x]$ of polynomials with integer coefficients is an integral domain. ■

■ **EXAMPLE 4** The ring $Z[\sqrt{2}] = \{a+b\sqrt{2} \mid a, b \in Z\}$ is an integral domain. ■

■ **EXAMPLE 5** The ring Z_p of integers modulo a prime p is an integral domain. ■

■ **EXAMPLE 6** The ring Z_n of integers modulo n is *not* an integral domain when n is not prime. ■

■ **EXAMPLE 7** The ring $M_2(Z)$ of 2×2 matrices over the integers is *not* an integral domain. ■

Field

- Let R be a ring with unity $1 \neq 0$. An element u in ring R is a unit in R if it has a multiplicative inverse in R.
- A commutative ring with unity where every non-zero element has a multiplicative inverse is called a **field**.
- The set of rational numbers $(\mathbb{Q})$ is a field because every non-zero rational number has a multiplicative inverse.
- However, the set of integers $(\mathbb{Z})$ is a commutative ring with unity, but it is not a field because multiplicative inverse does not exist.

Determine all the units in $\mathbb{Z}_{14}$. Verify whether $\mathbb{Z}_{14}$ forms a field.

$$\mathbb{Z}_{14} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}.$$

The elements of $\mathbb{Z}_{14}$ are the set of all integers modulo 14.

Inverse of $a \in \mathbb{Z}_{14}$ is b if $ab \equiv 1 \ (mod\ 14)$.

$$1^{-1} = 1, \quad 3^{-1} = 5, \quad 5^{-1} = 3, \quad 9^{-1} = 11, \quad 11^{-1} = 9, \quad 13^{-1} = 13$$

All other elements $\{0, 2, 4, 6, 7, 8, 10, 12\}$ do not have inverses.

$\mathbb{Z}_{14}$ is not a field.

An element $a \in \mathbb{Z}_n$ has a multiplicative inverse **if and only if**

$$\gcd(a, n) = 1.$$

$\mathbb{Z}_n$ is a field if and only if n is prime.

Question:

(a) Find all the units in $\mathbb{Z}_5$ and determine whether $\mathbb{Z}_5$ is a field.

(b) Find all the units in $\mathbb{Z}_4$ and determine whether $\mathbb{Z}_4$ is a field.

System	Elements	Units	Field?
$\mathbb{Z}_5$	{0,1,2,3,4}	{1,2,3,4}	✅ Yes
$\mathbb{Z}_4$	{0,1,2,3}	{1,3}	❌ No

Classify each nonzero element of $\mathbb{Z}_{20}$ as a unit or a 0 divisor.

Solution The greatest common divisor of m and 20 is 1 if $m = 1, 3, 7, 9, 11, 13, 17, 19$, so these are all units. For $m = 2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18$, $\gcd(m, 20) > 1$, so these are all 0 divisors. We see that

$$1 \cdot 1 = 3 \cdot 7 = 9 \cdot 9 = 11 \cdot 11 = 13 \cdot 17 = 19 \cdot 19 = 1 \in \mathbb{Z}_{20}$$

which verifies that each is a unit. We also see that

$$2 \cdot 10 = 4 \cdot 5 = 6 \cdot 10 = 8 \cdot 15 = 12 \cdot 5 = 14 \cdot 10 = 16 \cdot 5 = 18 \cdot 10 = 0 \in \mathbb{Z}_{20}$$

which verifies that each of these is a 0 divisor in $\mathbb{Z}_{20}$. ▲

Zero divisors

If a and b are two nonzero elements of a ring R such that $ab = 0$, then a and b are **divisors of 0** (or **0 divisors**). ■

- In $\mathbb{Z}_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$, the elements $2, 3, 4, 6, 8, 9, 10$ are divisors of 0.

  Note that these are exactly the numbers in $\mathbb{Z}_{12}$ that are not relatively prime to 12.

Theorem Let $m \in \mathbb{Z}_n$. Either $m = 0$, m is relatively prime to n, in which case m is a unit in $\mathbb{Z}_n$, or m is not relatively prime to n, in which case m is a 0 divisor in $\mathbb{Z}_n$.

Corollary If p is a prime number, then every nonzero element of $\mathbb{Z}_p$ is a unit, which means that $\mathbb{Z}_p$ is a field and it has no divisors of 0.

Divisors of Zero and Cancellation

Solve $x^2 - 5x + 6 = 0$.

The first thing we do is factor the left side:

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

Now we conclude that the only possible values for x are 2 and 3.

The reason is that if x is replaced by any number a, the product $(a - 2)(a - 3)$ of the resulting numbers is 0 if and only if either $a - 2 = 0$ or $a - 3 = 0$.

Solve $x^2 - 5x + 6 = 0$ in $\mathbb{Z}_{12}$.

$\mathbb{Z}_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

We have $x^2 - 5x + 6 = (x - 2)(x - 3)$.

In $\mathbb{Z}_{12}$, not only is $0a = 0 = a0, \forall\ a \in \mathbb{Z}_{12}$, but also

$$(2)(6) = (6)(2) = (3)(4) = (4)(3) = (3)(8) = (8)(3)$$
$$= (4)(6) = (6)(4) = (4)(9) = (9)(4) = (6)(6) = (6)(8)$$
$$= (8)(6) = (6)(10) = (10)(6) = (8)(9) = (9)(8) = 0.$$

Hence the equation has not only 2 and 3 as solutions, but also 6 and 11, for

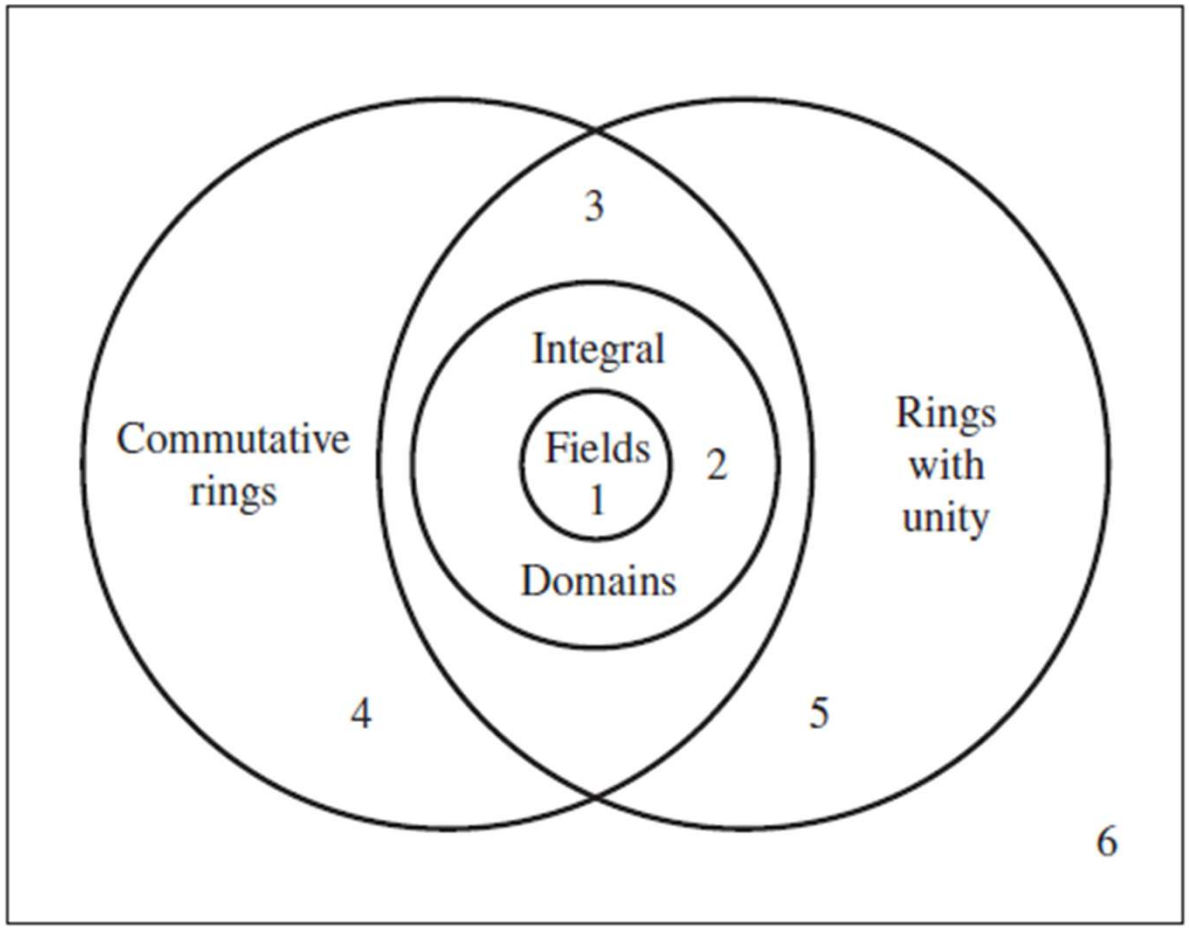
$(6 - 2)(6 - 3) = (4)(3) = 0$ and $(11 - 2)(11 - 3) = (9)(8) = 0$ in $\mathbb{Z}_{12}$.

Integral Domains

- An **integral domain** D is a commutative ring with unity $1 \neq 0$ that contains no divisors of 0.
  - Thus, if the coefficients of a polynomial are from an integral domain, one can solve a polynomial equation in which the polynomial can be factored into linear factors in the usual fashion by setting each factor equal to 0.
- $\mathbb{Z}$ and $\mathbb{Z}_p$ for any prime p are integral domains, but $\mathbb{Z}_n$ is not an integral domain if n is not prime.

The Characteristic of a Ring

- If for a ring R a positive integer n exists such that $n \cdot a = 0$ for all $a \in R$, then the least such positive integer is the characteristic of the ring R.
- If no such positive integer exists, then R is of characteristic 0.
- The ring $\mathbb{Z}_n$ is of characteristic n, while $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ all have characteristic 0.


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Rings
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unity
4
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6

Irreducible Polynomial

Suppose F is a field.

A polynomial $f(x) \in F[x]$ is **irreducible** if $f(x)$ cannot be written as a product of two polynomials $f_1(x), f_2(x)$, where $f_1(x)$ and $f_2(x)$ both have positive degree.

- Suppose that F is any finite field and suppose n is a positive integer. Then there is at least one irreducible polynomial of degree n in $(F[x], +, \cdot)$.

Example: In the ring $\mathbb{Z}_2[x]$,

- $x^2 + 1 = (x + 1)(x + 1)$, so $x^2 + 1$ is reducible.
- $x^2 + x = x(x + 1)$, so $x^2 + x$ is reducible.
- $x^2 + x + 1$ is irreducible.

Check the two elements of $\mathbb{Z}_2 = \{0, 1\}$:

- For $x = 0$: $0^2 + 0 + 1 = 1 \neq 0$.
- For $x = 1$: $1^2 + 1 + 1 = 1 + 1 + 1 = 3 \equiv 1 \pmod 2 \neq 0$.

Since $x^2 + x + 1$ has **no root** in $\mathbb{Z}_2$, it cannot factor as a product of linear factors in $\mathbb{Z}_2[x]$. Therefore it is **irreducible** in $\mathbb{Z}_2[x]$. □

Show that $f(x) = x^3 + 3x + 2$ viewed in $\mathbb{Z}_5[x]$ is irreducible over $\mathbb{Z}_5[x]$.

Compute $f(x) = x^3 + 3x + 2$ modulo 5:

- $x = 0: \ f(0) = 0^3 + 3 \cdot 0 + 2 = 2 \not\equiv 0 \pmod 5$.
- $x = 1: \ f(1) = 1 + 3 + 2 = 6 \equiv 6 - 5 = 1 \not\equiv 0 \pmod 5$.
- $x = 2: \ f(2) = 8 + 6 + 2 = 16 \equiv 16 - 15 = 1 \not\equiv 0 \pmod 5$.
- $x = 3: \ f(3) = 27 + 9 + 2 = 38 \equiv 38 - 35 = 3 \not\equiv 0 \pmod 5$.
- $x = 4: \ f(4) = 64 + 12 + 2 = 78 \equiv 78 - 75 = 3 \not\equiv 0 \pmod 5$.

No element of $\mathbb{Z}_5$ is a root, so $f(x)$ has no linear factor in $\mathbb{Z}_5[x]$. Hence $f(x)$ cannot factor nontrivially (the only possible factorization of a cubic over a field would include a linear factor), so

$$\boxed{x^3 + 3x + 2 \text{ is irreducible in } \mathbb{Z}_5[x].}$$

Finite Fields

- There exists a finite field of order q if and only if $q = p^n$ where p is prime and n $n \geq 1$ is an integer.
- Because there is only one field for each prime-power p^n, we may unambiguously denote it by $GF(p^n)$, and call it the Galois field of order p^n.

Structure of Finite Fields

As a group under addition, $GF(p^n)$ is isomorphic to

$$\underbrace{Z_p \oplus Z_p \oplus \ldots \oplus Z_p}_{n \text{ factors}}.$$

As a group under multiplication, the set of nonzero elements of $GF(p^n)$ is isomorphic to Z_{p^n-1} (and is, therefore, cyclic).

Additive Structure

- The elements of a finite field $GF(p^n)$ form a group under addition.
- $(GF(p^n), +) \cong \mathbb{Z}_p \oplus \mathbb{Z}_p \oplus \mathbb{Z}_p \oplus \cdots \oplus \mathbb{Z}_p$
- Every element of $GF(p^n)$ can be represented as an n −tuple over $\mathbb{Z}_p$.

Multiplicative Structure

- The nonzero elements of a finite field $GF(p^n)$ form a group under multiplication.
- $(GF(p^n)^*, \times) \cong \mathbb{Z}_{p^n-1}$
- This means there exists an element g (called a primitive element) such that every nonzero element of $GF(p^n)$ can be written as some power of g.

Determine whether there exists a finite field having the given number of elements:

1) 4096

2) 3127

3) 68,921

If a finite field exists, describe its structure. Further, if α is a primitive element of the field, find the order of α and determine how many primitive elements the field contains.

- A finite field of order q exists **iff** $q = p^n$ for some prime p and integer $n \geq 1$.
- If α is a **primitive element** (generator of the multiplicative group), then:
  - The **order** of α is $q - 1$.
  - The **number of primitive elements** is $\varphi(q-1)$ (Euler's totient function), since primitive elements are the generators of the cyclic group $\mathbb{F}_q^\times$ of order $q - 1$

1) $4096 = 2^{12} \Rightarrow$ a finite field exists.

Additive group: $(\mathbb{F}_{4096}, +) \cong \mathbb{Z}_2^{12}$,
Multiplicative group: $(\mathbb{F}_{4096}^{\times}, \cdot) \cong \mathbb{Z}_{4095}$.

If α is a primitive element, $order(\alpha) = 4096 - 1 = 4095$.

Now, $4095 = 3^2.5.7.13$

Number of primitive elements = $\phi(4095)$

$$= 4095\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{7}\right)\left(1 - \frac{1}{13}\right)$$

$$= 1728$$

2) $3127 = 53 * 59$, *not a prime power*

$\Rightarrow$ no finite field of order 3127 exists.

3) $68921 = 41^3 \Rightarrow$ a finite field exists.

Additive group: $\mathbb{Z}_{41} \oplus \mathbb{Z}_{41} \oplus \mathbb{Z}_{41}$,
Multiplicative group: $\mathbb{Z}_{68,920}$.

If α is a primitive element, $order(\alpha) = 68921 - 1 = 68920$.

Now, $68920 = 2^3.5.1723$

Number of primitive elements = $\phi(68920)$

$$= 68920\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{1723}\right)$$

$$= 27552$$

Questions

1. In $GF(17)$, compute the multiplicative inverse of 7 using the extended Euclidean algorithm.
2. Find the inverse of 11 in $GF(19)$. Verify by multiplication.
3. Using the extended Euclidean algorithm, compute the inverse of 13 in $GF(23)$.
4. Find all zero divisors in $\mathbb{Z}_{14}$. List each zero divisor a and a nonzero b such that $ab \equiv 0(mod 14)$.
5. ElGamal parameters: $p = 23,\ g = 5$. Receiver's private key $x = 6$. Sender wants to send $M = 13$ and chooses random $k = 4$. Compute the ciphertext pair (C_1, C_2) and show how the receiver recovers M.

1. Receiver's public key

$$h = g^x \bmod p = 5^6 \bmod 23.$$

Compute powers: $5^2 = 25 \equiv 2,\ 5^4 = (5^2)^2 = 2^2 = 4.$

Then $5^6 = 5^4 \cdot 5^2 = 4 \cdot 2 = 8.$

So $h = 8.$

2. Sender computes ciphertext (C_1, C_2)

$$C_1 = g^k \bmod p = 5^4 \bmod 23 = 4.$$

$$C_2 = M \cdot h^k \bmod p.$$

Compute $h^k = 8^4$. $8^2 = 64 \equiv 18$, so $8^4 = (8^2)^2 = 18^2 = 324 \equiv 2 \pmod{23}$.

Thus $C_2 = 13 \cdot 2 \bmod 23 = 26 \bmod 23 = 3.$

So the ciphertext is $\boxed{(C_1, C_2) = (4,\ 3)}$.

3. Receiver decrypts

Receiver computes shared secret $s = C_1^x \bmod p = 4^6 \bmod 23$.

$4^2 = 16,\ 4^3 = 64 \equiv 18$, so $4^6 = (4^3)^2 = 18^2 = 324 \equiv 2$. Thus $s = 2$.

Find s^{-1} mod 23: $2 \cdot 12 = 24 \equiv 1$, so $s^{-1} = 12$.

Recover message:

$$M = C_2 \cdot s^{-1} \bmod p = 3 \cdot 12 \bmod 23 = 36 \bmod 23 = 13.$$

So decryption returns $\boxed{M = 13}$, as required.

Partially Ordered Sets

A relation $\leq$ on a set X is called a partial order on X if and only if for every $x, y, z \in X$ the following conditions are satisfied:

i. $x \leq x$ (reflexive)

ii. $x \leq y$ and $y \leq x \Rightarrow x = y$ (antisymmetric)

iii. $x \leq y$ and $y \leq z \Rightarrow x \leq z$ (transitive)

The inverse relation of $\leq$, denoted by $\geq$, is also a partial order on X.

Lattices

A lattice is a partially ordered set L such that every subset $\{x, y\}$ consisting of two elements has a least upper bound (lub) and a greatest lower bound (glb).

If L is a lattice, then we define

- $x \wedge y = glb\{x, y\}$
- $x \vee y = lub\{x, y\}$

Example 1:

- *Set:* Power set of a set, partially ordered by **inclusion**
- *Supremum:* **Union** of subsets
- *Infimum:* **Intersection** of subsets

Example 2:

- *Set:* Natural numbers, partially ordered by **divisibility**
- *Supremum:* **Least Common Multiple (LCM)**
- *Infimum:* **Greatest Common Divisor (GCD)**

Example 1: Power Set Ordered by Inclusion

- **Set:** Consider a set S. Its **power set**, denoted by $\mathcal{P}(S)$, is the set of all subsets of S.
  Example: If $S = \{a, b\}$, then
  $\mathcal{P}(S) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.
- **Partial Order:** We define a relation "$\subseteq$" (subset or inclusion) on $\mathcal{P}(S)$.
  That is, for any $A, B \subseteq S$, we say $A \leq B$ if $A \subseteq B$.
  This relation is **reflexive**, **antisymmetric**, and **transitive**, so it forms a **partially ordered set (poset)**.
- **Supremum (Least Upper Bound):**
  The supremum of two subsets A and B is their **union** $A \cup B$.
  This is the smallest set that contains both A and B.
- **Infimum (Greatest Lower Bound):**
  The infimum of two subsets A and B is their **intersection** $A \cap B$.
  This is the largest set contained in both A and B.

Example 2: Natural Numbers Ordered by Divisibility

- **Set:** The set of **natural numbers** $\mathbb{N} = \{1, 2, 3, 4, \dots\}$.
- **Partial Order:**

  We define a relation "divides" (|) on $\mathbb{N}$.

  For any $a, b \in \mathbb{N}$, we say

  $a \leq b$ if a **divides** b (that is, $b = ka$ for some integer k).

  This relation is **reflexive**, **antisymmetric**, and **transitive**, so it forms a **partially ordered set (poset)**.
- **Supremum (Least Upper Bound):**

  The supremum of two numbers a and b (with respect to divisibility) is their **Least Common Multiple (LCM)**.

  → It is the smallest number that both a and b divide.
- **Infimum (Greatest Lower Bound):**

  The infimum of a and b is their **Greatest Common Divisor (GCD)**.

  → It is the largest number that divides both a and b.

Sublattice

A sublattice of a lattice L is a subset S of L that is closed under meet and join operations of L.

For all $a, b \in S$,

- $a \vee b \in S$ (join)
- $a \wedge b \in S$ (meet)

Every sublattice is itself a lattice under the same partial order.

Ex: If $L = (P(S), \subseteq)$, then the set of all subsets of S containing a fixed element forms a sublattice of L.

Complete Lattice

A lattice L is said to be complete if and only if for each of its subsets X (not just pairs of elements), $\inf X$ and $\sup X$ exist.

We define the symbols $\bigwedge X = \inf X$ and $\bigvee X = \sup X$.

Example 1:

- The **power set** of a set, ordered by **inclusion (⊆)**
- **Supremum:** Union of subsets
- **Infimum:** Intersection of subsets
- Forms a **complete lattice**, since union and intersection exist for all subsets

Example 2:

- The **set of natural numbers**, ordered by **divisibility (|)**
- **Supremum:** Least Common Multiple (LCM)
- **Infimum:** Greatest Common Divisor (GCD)
- Forms a **lattice but not a complete lattice**, since some subsets (e.g., all natural numbers) have no LCM in $\mathbb{N}$

Bounded Lattice

A lattice L is said to be bounded if it has both a greatest element (1) and a least element (0).

Example:

- The **power set** of a set S, ordered by inclusion ($\subseteq$).
  - **0:** $\emptyset$ (empty set)
  - **1:** S (the whole set)

Complemented Lattice

A bounded lattice in which every element a has a complement b such that

$$a \vee b = 1 \text{ and } a \wedge b = 0.$$

Example:

- In the **power set lattice** $\mathcal{P}(S)$:
  - The complement of subset A is $A' = S - A$.
  - $A \cup A' = S$, and $A \cap A' = \emptyset$.

Distributive Lattice

A lattice where the operations meet and join are distributive over each other:

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c).$$

Example:

- The **power set lattice** $\mathcal{P}(S)$ under inclusion ($\subseteq$).
  - **Join:** Union
  - **Meet:** Intersection
  - Set union and intersection satisfy distributive laws.